CHAPTER

# 11 Dual Nature of Radiation and Matter

## Planck's Quantum Theory

- According to Planck's quantum theory, light is considered to be made up of small packets (or particles) of energy known as quanta of energy or radiation.

  Energy, $E = h\nu = \dfrac{hc}{\lambda} = \dfrac{12400}{\lambda(\text{Å})}$ eV

## Photons

- Momentum of one photon is $\dfrac{h}{\lambda}$.
- When radiation interacts with matter, the radiation behaves as if it is made of particles like photons.
- Einstein proposed that electromagnetic radiation (or simply light) is quantized and exists in elementary amounts (quanta) that we now call photons.
- Photons are not deflected by electric and magnetic fields which shows that they are neutral and do not carry any charge.
- The energy of photon depends upon the frequency of radiation but is independent of the intensity of radiation.

## Photoelectric Effect

- When light of suitable frequency illuminates a metal surface, electrons are emitted. This process of ejection of electrons using light is known as photoelectric emission. Photoelectrons ejected from metal have kinetic energies ranging from 0 to $K_{max}$.
- A certain minimum amount of energy is required for an electron to be pulled out from the surface of a metal. This minimum energy is called the work function ($\phi$) of that metal. Work function is minimum for cesium (1.9 eV).
- Einstein equation for photoelectric effect is,

  $$h\nu = \phi + KE_{max} \quad \Rightarrow \frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV_s$$

- The minimum frequency of the incident light below which photoelectrons are not ejected from the metal surface is known as threshold frequency ($\nu_0$).

  Work function, $\phi = h\nu_0 = \dfrac{hc}{\lambda_0}$

  $\lambda_0$ = threshold wavelength
- The minimum negative potential given to the metal plate with respect to the collector at which the photoelectric current becomes zero is known as stopping potential or cut-off potential. Here $KE_{max} = eV_s, V_s =$ stopping potential

  Stopping potential is independent of intensity of light used.
- The number of photoelectrons emitted per second is directly proportional to the intensity of the incident radiation.
- The maximum kinetic energy of the ejected electrons is independent of the intensity of incident radiation but depends upon the frequency of the incident radiation.

## Radiation Pressure

Radiation pressure, $P = \dfrac{I}{c}(1+r)$. Here $I$ is the intensity of incident radiation, $c$ is the speed of light and $r$ is the reflectivity of the surface.

For 100% reflection, $r = 1$ and for 100% absorption $r = 0$.

## de-Broglie Hypothesis

- It says that a wave is associated with a moving material particle. The wavelength associated with a moving particle is given by $\lambda = \dfrac{h}{mv}$, where $m$ is the mass of the particle moving with $v$ velocity and h is Planck's constant. This wave is called de-Broglie wave.

### Key Tips

- de-Broglie wavelength of a material particle, $\lambda = \dfrac{h}{mv}$
- de-Broglie wavelength in terms of energy of a particle ($E$), $\lambda = \dfrac{h}{\sqrt{2mE}}$
- de-Broglie wavelength of an electron accelerated through a potential $V$ volt, $\lambda = \sqrt{\dfrac{150}{V}}$ Å $= \dfrac{12.27}{\sqrt{V}}$ Å
- de-Broglie wavelength of a particle in terms of temperature ($T$), $\lambda = \dfrac{h}{\sqrt{3mkT}}$

## Heisenberg's Uncertainty Principle

According to Heisenberg's Uncertainty Principle, it is not possible to measure exactly both the position and momentum of a microscopic particle (say electron) at the same time. That is,

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \text{ where } \hbar = \frac{h}{2\pi},$$